

Anti-epileptics Antidepressants and Antipsychotics

Learning Outcomes

**By the end of this lecture,
students should be able to:**

Describe antiepileptics

Outline antidepressant drug therapy

Describe Antipsychotics

DRUG TREATMENT OF EPILEPSIES

Anti-epileptics

Definition

- **Anti-epileptic** drugs inhibit the spread of abnormal electrical discharge in the brain with minimal general depressant action on the central nervous system.

Drugs used in epilepsies

- **Older AEDs:** Examples like **Phenytoin, Carbamazepine, and Valproate.**
- **Newer AEDs:** Examples like **Levetiracetam**

PHENYTOIN



Mechanism:

Phenytoin prevents repetitive depolarization of the normal brain cells and stabilizes neuronal membranes.



Kinetics: After oral administration peak plasma concentration of phenytoin usually takes 2 to 4 hours.



The majority of the dose is excreted in the urine.

Phenytoin



Adverse effects: **gum hypertrophy**, hirsutism, hypersensitivity reaction, megaloblastic anemia and hyperglycemia.



Uses: Prophylactic treatment of all varieties seizures



It is also used in prophylactic treatment of generalized convulsive seizures and treatment of status epilepticus.

CARBAMAZEPINE



Similar to
phenytoin



Uses: It is
effective in
grandmal
(tonic-clonic)



Poor water
solubility has
slow oral
absorption.



Kinetics:
Metabolized in
the liver to an
active
metabolite
(epoxy
carbamazepin
e)



**Adverse
effects:**
Sedation,
dizziness and
extrapyramida
l side effects,
dry mouth,
and blurred
vision.

SODIUM VALPROATE (VALPROIC ACID)



Mechanism: Sodium valproate increases the levels of GABA in the brain.



It is therefore sometimes called the broad-range antiepileptic drug.



Adverse effects include anorexia, nausea, vomiting, diarrhea, and/or constipation.

Key Information for Dentists

- **Phenytoin:** Commonly used; can cause gingival hyperplasia.
- **Carbamazepine:** Known for drug interactions; induces liver enzymes.
- **Valproate:** Risk of bleeding due to platelet suppression.
- **Levetiracetam:** Fewer drug interactions; widely used in dental patients.

Xerostomia (Dry Mouth): Common with various AEDs, leading to increased caries risk.

Antidepressants

Anti Depressants

- **Depression:** Feelings of hopelessness, lack of energy (or feeling "weighed down"), and taking little or no pleasure in things that gave joy in the past.
- **Types of antidepressant drugs**
 - Tricyclic antidepressant (TCA): Imipramine.
 - Selective serotonin (5-HT) uptake inhibitors:
Fluoxetine
 - Monoamine Oxidase Inhibitors (MAOIs):
Phenelzine
 - Atypical Antidepressants:
Bupropion

Imipramine

Mechanism: Block the amine uptake (NE and 5-HT).

Pharmacologic effects:

- (1) CNS: a non-depressed person experiences sleeping. In the depressed patient, an elevation of mood occurs 2-3 weeks after administration begins.
- (2) Autonomic nervous system: anticholinergic effects.
- (3) Cardiovascular effects: orthostatic hypotension and arrhythmias.

Fluoxetine

Mechanism of action:

- (1) a selective inhibitor of serotonin uptake in the CNS.
- (2) little effect on central norepinephrine and dopamine function.
- (3) less adverse effects because of minimal binding to cholinergic, histaminic, and α -adrenergic receptors.

- **Therapeutic uses:**

- Used for treatment of mild to moderate endogenous depression and obsessive-compulsive disorder, obesity

- **Adverse effects:**

- (1) Cause anorexia.
- (2) Cause mania
- (3) Result in nausea, nervousness, headache, and insomnia.

Effects of teeth grinding (Bruxism)



Oral Health Side Effects of Antidepressants

Bruxism: Often seen with SSRIs; can cause tooth wear.

Taste Alterations: Changes in taste perception, particularly with TCAs and MAOIs.

Delayed Healing and Bleeding Risks: Antidepressants affecting serotonin can alter platelet function.

ANTIPSYCHOTICS

Schizophrenia affects about 1% of the population.

It affects young people, is often chronic and usually highly disabling.

There is a strong hereditary factor.

Psychosis is a disorder characterized by disturbances of reality and perception, impaired cognitive functioning, and inappropriate or diminished mood.

The main clinical features

Positive symptoms:

- Delusions (false belief)
- Hallucinations, usually in the form of voices, and often exhortatory in their message
- Thought disorder, abnormal behaviors, such as stereotyped or occasionally aggressive behaviors.

Negative symptoms:

- withdrawal from social contacts
- flattening of emotional responses.

In addition, deficits in cognitive function (e.g. attention, memory) are often present, together with anxiety and depression, leading to suicide in about 10% of cases.

Pathogenesis

Dopamine theory:

- Increased dopamine levels may be the cause of the disease.
- In animals, **dopamine** release causes schizophrenic symptoms.
- Potent D2-receptor agonists (**e.g. apomorphine**) produce similar effects in animals.

Classification of antipsychotic drugs

classical 'typical' antipsychotics
(e.g. chlorpromazine, haloperidol)

recent 'atypical' antipsychotics
(e.g. clozapine, risperidone)

The incidence of extrapyramidal
(Movement disorders) side effects is
less in the atypical group.

- The efficacy in treatment-resistant groups of patients and efficacy against negative symptoms are the main difficulties.

Mechanism of action of antipsychotic drugs

- All antipsychotic drugs are antagonists at dopamine D₂ receptors but most also block other receptors, especially 5-HT₂.
- Antipsychotic potency is due to activity on D₂-receptors, but other activities may determine side-effect profile.
- Antipsychotics take days or weeks to work.
- **Clozapine** also blocks D₄-receptors.

Antipsychotic-induced motor disturbances

Major problem of antipsychotic drug treatment.

Parkinson-like symptoms(Acute dystonia)

slowly developing tardive dyskinesia, often irreversible.

Acute symptoms comprise involuntary movements, of tremor and rigidity.

Incidence of acute dystonias and tardive dyskinesia is less with atypical antipsychotics, and particularly low with **clozapine**.

Other side effects

Dry mouth, blurred vision, hypotension

With clozapine, leucopenia is common and requires routine monitoring.

Antipsychotic malignant syndrome is a rare but potentially dangerous.

Dental Implications

Complaints and not cooperative many times

Artificial salivary products, antiseptic mouthwash, and a 0.05% sodium fluoride mouth rinse are prescribed for many patients with signs of Xerostomia.

Application of fluoride gel, and dental restorations.

Profound local anesthesia is mandatory to adequately perform these procedures in depressed and often anxious individuals.

Dental Implications Contd'

Be aware of the possibility that your patient may be phobic or distrustful.

Learn to recognize signs of fear, and when in doubt, ask your patient in an accepting way about possible fear.

Numerous questions or comments on prior dental experiences may be a sign of distrust.

Explain procedures and their rationale, and obtain the patient's permission before doing each procedure.

For some patients, explanation and permission may be advisable even for actions such as lowering the back of the dental chair. It is much more important that the patient return for the next appointment than it is to complete a set amount of dental work at the first appointment.

References

- Bart –Jhonson, Frank J. Dowd. ***Pharmacology and Therapeutic for Dentistry***, 6th edition, 2011. Elsevier Publishers,USA
- Karen Whalen, Richard Finkel, Thomas A Panavelil. Lippincott Illustrated Reviews Pharmacology. 6th ed. 2015. Philadelphia Wolter Kluwer Puplisher.